

Problem

An oscillogram can be adequately modeled by a second-order system in the form of a parallel RLC circuit. It is desired to give an underdamped voltage across a $200\text{-}\Omega$ resistor. If the damped frequency is 4 kHz and the time constant of the envelope is 0.25 s , find the necessary values of L and C .

Step-by-step solution

Step 1 of 1

$$t = \frac{1}{\alpha} = 0.25 \quad \text{Leads } 2 = 4$$

$$\alpha = \frac{1}{2RC}$$

$$\text{Or } C = \frac{1}{2\alpha R} = \frac{1}{2 \times 4 \times 200} = 625 \mu F$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$\omega_0^2 = \omega_d^2 + \alpha^2 = (2\pi \times 4 \times 10^3)^2 + 16$$

$$= (2\pi \times 4 \times 10^3)^2 = \frac{1}{LC}$$

$$\Rightarrow L = \frac{1}{64\pi^2 \times 10^6 \times 625 \times 10^{-6}}$$

$$= 2.5333 \mu F$$